

FL-Iroh: NAT-Transparent Federated Learning for Agricultural and Environmental IoT via QUIC P2P Transport and CoAP Service Discovery

Raúl López-Blanco^{a,d,*}, Ricardo S. Alonso^{b,d}, Tiago Pinto^c and Javier Prieto^a

^aBISITE Research Group, University of Salamanca, Edificio I+D+i, Calle del Espejo 2, Salamanca, 37006, Castile and León, Spain

^bAIR Institute – Deep tech lab, IoT Digital Innovation Hub, Paseo de Belén 9A, Valladolid, 47011, Castile and León, Spain

^cUniversity of Trás-os-Montes and Alto Douro, Vila Real, Portugal

^dUNIR, International University of La Rioja, Av. de la Paz, 137, Logroño, 26006, La Rioja, Spain

ABSTRACT

Author Contributions

Raúl López-Blanco (Corresponding Author)

ORCID: 0000-0002-8856-4008 | Email: raul1b@usal.es

Affiliations:

- BISITE Research Group, University of Salamanca, Spain
- UNIR, International University of La Rioja, Spain

Contributions: Conceptualization, Methodology, Software, Investigation, Data curation, Writing – original draft

Ricardo S. Alonso

ORCID: 0000-0002-6599-0186 | Email: ralonso@air-institute.com

Affiliations:

- AIR Institute – Deep tech lab, IoT Digital Innovation Hub, Valladolid, Spain
- UNIR, International University of La Rioja, Spain

Contributions: Conceptualization, Methodology, Writing – review & editing, Supervision

Tiago Pinto

ORCID: 0000-0001-8248-080X | Email: tiago.m.pinto@gmail.com

Affiliation: University of Trás-os-Montes and Alto Douro, Vila Real, Portugal

Contributions: Writing – review & editing, Supervision

Javier Prieto

ORCID: 0000-0001-8175-2201 | Email: javierp@usal.es

Affiliations:

- BISITE Research Group, University of Salamanca, Spain

Contributions: Writing – review & editing, Supervision, Funding acquisition

Data and Code Availability

Source code, experiments, results, and processed datasets are available at:

<https://github.com/raul1b34/FL-Iroh>

Supplementary Materials

The following supporting materials are available with this manuscript:

- Crop Recommendation dataset (publicly available on Kaggle)
- Raw air-quality data from Junta de Castilla y León
- Meteorological data from AEMET (Spanish State Meteorological Agency)
- Experimental harness and per-run results backing all tables

Acknowledgements

This work has been supported by the OSTARA project under the Partnership for Research and Innovation in the Mediterranean Area (PRIMA), the MICIU/AEI/10.13039/501100011033 and the European Union (grant number PCI2026-177466-1).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

*Corresponding author

✉ raul1b@usal.es (R. López-Blanco); ralonso@air-institute.com (R. S. Alonso); tiago.m.pinto@gmail.com (T. Pinto); javierp@usal.es (J. Prieto)
ORCID(s): 0000-0002-8856-4008 (R. López-Blanco);
0000-0002-6599-0186 (R. S. Alonso); 0000-0001-8248-080X (T. Pinto);
0000-0001-8175-2201 (J. Prieto)